

T-740-SPMM Final Project: Medical Summary

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Executive Summary

In this phase, the team focused on creating a working prototype of the application based on the results of previous iterations. Initially, the team selected three out of forty-five user stories, representing the minimum set of features they considered necessary for the prototype. Then, they planned the Sprints by dividing all of the user stories in the product backlog into twelve Sprints based on priority. For the first Sprint, the team used a Trello board to organize the work.

After the planning stage, the team began the design phase. They defined the design approach to be used and identified four tasks for user testing. For each task, they described the happy path. To validate the prototype, user testing was conducted, and the results were analyzed to gain a better understanding of user needs and gather feedback to improve the planning of future Sprints.

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Chapter 1: Introduction

This final report summarizes the team's iterative development process throughout the course. In this phase, the team refined its previous work and built upon it to complete the first Sprint, creating a prototype and performing user testing.

During previous project steps, the team held introductory briefings with stakeholders to clarify objectives and establish a shared understanding of the project's intended impact. The team also conducted initial research to identify key user groups and held user interviews to better understand user requirements and expectations. Based on these findings, the team submitted a formal project proposal that outlined the product backlog, potential risks, and stakeholder needs and resources.

This phase covered the reality check of the UCD Sprint. Using the artifacts produced in previous phases, the team created a product prototype and conducted user testing. The team documented the activities performed during this phase, as outlined below:

- **User stories:** The team selected the most important user stories representing the minimum viable features for the prototype. They organized the Sprints using a Trello board.
- **Product backlog:** The team reviewed and updated the product backlog. They planned future Sprints by assigning user stories to specific iterations and included the previously selected stories in the first Sprint.
- **Prototyping and Testing:** The team outlined the design process, user tasks, and their corresponding happy paths.
- **Analysis:** The team analyzed the obtained results to understand user behavior and identify areas for improvement.
- **Stakeholder meeting:** The team verified that the project was progressing in the right direction, collected feedback, and identified issues or additional requirements.

Finally, the team presented their conclusions on the overall process and its implications for ongoing development.

Chapter 2: User Stories

For the development of our project we are following the Agile software development methodology and structuring our work into Sprints. The first Sprint focuses on establishing the core functionality of the system, the bare minimum that demonstrates the main idea of our product.

When prioritizing user stories from our product backlog, we considered which features are essential for the basic use of our system. Since the main goal of our application is to allow Icelandic travelers to access and present their medical summaries abroad, we determined that the following user stories are essential for the application to be considered functionally complete:

- As an Icelandic traveler, I want to create a medical summary so that I can travel with it.
- As an Icelandic traveler I want to choose the medical information that is on my summary so that I can show relevant information.
- As an Icelandic traveler I want to change the language of my medical summary so that I can travel with it abroad.
- As an Icelandic traveler I want to download my medical summary so that I can travel with it and look it up offline.
- As an Icelandic traveler I want to ensure my healthcare information is certified so that the foreign healthcare professional can easily verify its authenticity.
- As an Icelandic traveler I want to log out easily so that no one else can use my device to view my data.
- As an Icelandic traveler I want my offline downloaded file encrypted so that no one can read it if my phone is lost.
- As a doctor abroad I want to see the translated medical summary so that I can treat the patient properly.
- As an Icelandic traveler I want to be able to open my summary with fingerprint or face ID so that it's quick and secure.
- As an Icelandic traveler I want the option to delete my offline copy so that no traces remain on my device.

From these, we identified three critical user stories to be implemented during Sprint 1, forming the foundation of the Minimum Viable Product (MVP):

1. As an Icelandic traveler, I want to create a medical summary so that I can travel with it.

2. As an Icelandic traveler, I want to choose the medical information that is on my summary so that I can show relevant information.
3. As an Icelandic traveler, I want to change the language of my medical summary so that I can travel with it abroad.

These core functionalities serve as the foundation for implementing additional features such as downloading, encryption, and certification. By focusing on these, we ensure that Sprint 1 delivers a usable and testable version of the product that later can be expanded in future Sprints.

We decided that each Sprint would last two weeks, which gives enough time to complete tasks while still allowing feedback and adaption. The Sprint was organized using Trello, where each user story was divided into smaller tasks and tracked through work stages such as To Do, In Progress, Testing and Done.

Work breakdown for the first Sprint:

<https://trello.com/invite/b/68ff8f90868698c4925fd11e/ATTI6e0567b42e909cc91a9bb697c5bc9c02445B5AD2/spmm-final-project>

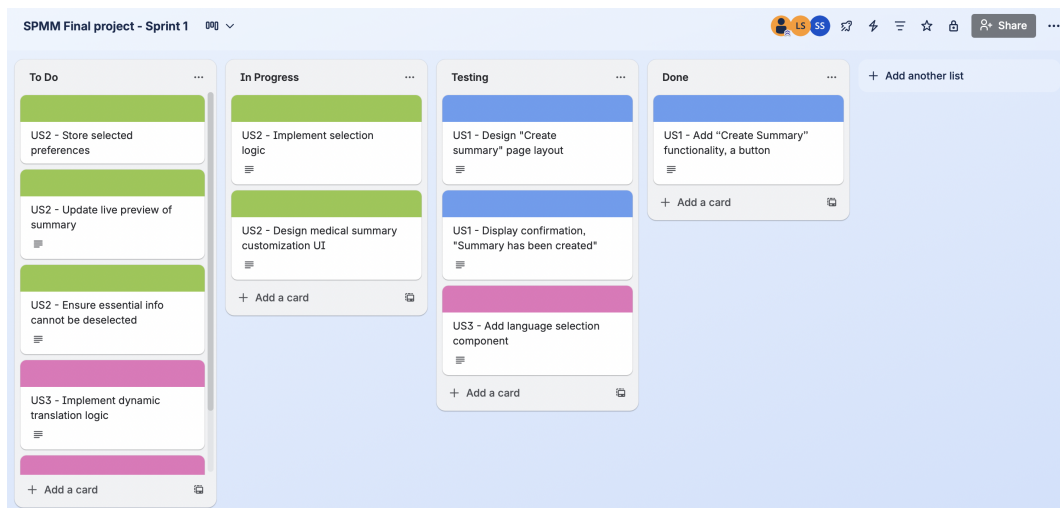


Figure 2.1: Trello board for Sprint 1 showing user stories and progress

Chapter 3: Product backlog

The product backlog includes all the user stories created by the team. These stories were divided into sprints to plan and organize the development process. Each sprint lasts two weeks, giving the team time to make progress and receive feedback. Some user stories may not be fully completed within one sprint, in which case the remaining work will be carried over and finished in the next sprint.

We structured the sprints so that higher priority user stories are addressed first, followed by those with lower priority. Additionally, dependencies between tasks were carefully considered to ensure prerequisite stories were completed before those that rely on them. In total, the backlog includes 45 user stories, however not all are intended to be completed, as some lower priority stories were documented primarily as a potential future enhancements or ideas.

To justify the structure of the sprint plan, the first three Sprints focus on the core functionality needed for a usable MVP: creating a medical summary, selecting its contents, and translating it. These features are essential for the main purpose of the application and must exist before any additional functionality can be tested or evaluated. Once this core flow is in place, the following sprints introduce supporting features such as downloading, PDF export, and medical data warnings, which depend on the summary being created first. Security-related stories (e.g., encryption, authentication, biometric access) are scheduled later because they require a working export mechanism and involve greater technical complexity. Finally, low-impact enhancements such as dark mode, tutorial guides, or automatic country detection are placed in the last Sprints, as they do not affect the main medical use case and can be added independently.

Name: Short name to describe the user story.

Description: User story; As a <user>, I want to <do something>, so that I can <some reason>.

Story Points: S = Less than 1 week of development, M = 1-2 weeks of development, L = 3-4 weeks of development, XL = More than 4 weeks of development.

Acceptance Criteria: Conditions that must be met for the user story to be considered done.

Sprint: In which Sprint we think this will be handled.

ID	Name	Description	Story Points	Acceptance Criteria	Sprint
1	Create	As an Icelandic traveler, I want to create a medical summary so that I can travel with it.	L	User can enter valid <i>Ísland.is</i> credentials and granted access. After successful login, the user goes into health, medical summary and creates their summary.	Sprint 1
2	Choosing information	As an Icelandic traveler I want to choose the medical information that is on my summary so that I can show relevant information.	M	User can select/deselect items to have on summary. Vital items show a warning if deselected. On Save, the summary is displayed. If saving fails, the user sees a clear error and previous setting remain unchanged.	Sprint 1
3	Warning	As an Icelandic traveler, I want the system to warn me before I remove any vital medical information from my summary, so that I don't accidentally delete critical details that a healthcare professional might need.	S	When the user selects to remove vital information, the system displays a warning message. The user must confirm or cancel the removal. If the user confirms, the info is removed, and summary is updated. If the user cancels, the info remains in the summary unchanged. If the info is non vital, the system removes it without showing a warning.	Sprint 6
4	Changing language	As an Icelandic traveler I want to change the language of my medical summary so that I can travel with it abroad.	L	User can choose a language from a list of supported languages. Once selected, the medical summary is displayed immediately in that language. If translation fails, the user sees a clear error message and the last available version remains visible.	Sprint 1

ID	Name	Description	Story Points	Acceptance Criteria	Sprint
5	Download summary	As an Icelandic traveler, I want to download my medical summary so that I can travel with it and look it up offline.	S	User can download their summary. After downloading, the summary is stored securely on the device and accessible without internet. The app displays a confirmation message when successful. The summary is readable offline. If download fails, the user sees a clear error message and can retry.	Sprint 2
6	Travel with summary	As an Icelandic traveler, I want to travel with a medical summary so that if something might happen, I have an up-to-date summary for the healthcare professionals.	S	User can download their medical summary from <i>Ísland.is</i> . The summary includes up-to-date information. The summary is available offline once downloaded. User receives a confirmation message after successful download.	Sprint 6
7	Simple and easy	As an Icelandic traveler with a low digital literacy, I want simple and easy steps in downloading it so that I can do it on my own or with minimal assistance.	S	The downloading option is clearly visible. The download process takes no more than 2-3 steps. Each step includes clear instructions with simple wording and/or icons. The system provides a confirmation message once the download is complete. If something goes wrong, the user receives a clear error message with guidance.	Sprint 2

ID	Name	Description	Story Points	Acceptance Criteria	Sprint
8	Changing language on the fly	As an Icelandic traveler, I want to be able to change the language of my medical summary on the fly so that doctors abroad can understand it even if it is not originally in their language.	L	User can select a different language for their summary from a list of supported languages. When selected, the summary is displayed in the chosen language immediately. If translation fails, the user sees an error message and the last successfully displayed version remains visible.	Sprint 3
9	Real time update	As an Icelandic traveler, I want my medical information to update in real time whenever a doctor adds new information so that I can always have the medical summary up to date.	L	When a doctor adds/changes information, the user's medical summary in <i>Ísland.is</i> is updated within 1-2 minutes. The user gets a notification when new information has been added. The user can manually refresh their summary to check for new updates.	Sprint 6
10	Fast translation	As an Icelandic traveler, I want the translation to be fast so that I can quickly show doctors my summary.	L	When the user requests a translation, the summary is displayed in the selected language within 2 seconds. If translation fails, the user sees a clear error message and can retry.	Sprint 7
11	Certified information	As an Icelandic traveler, I want to ensure my healthcare information is certified so that the foreign healthcare professional can easily verify its authenticity.	M	The medical summary includes a digital certificate or seal issued by <i>Ísland.is</i> . Foreign doctors can see the certification mark on the document. The certification can be verified electronically. If the certification is invalid, the system clearly shows that.	Sprint 7

ID	Name	Description	Story Points	Acceptance Criteria	Sprint
12	Remove form summary	As an Icelandic traveler, I want to remove information on my medical information so that I can have what I want on the summary.	S	User can select and remove an item on the summary. The system updates the summary to no longer show the removed item. If the removed information is considered critical, the system displays a warning before final removal. User can confirm or cancel the removal after seeing the warning.	Sprint 3
13	Two factor authentication	As an Icelandic traveler, I want two factor authentication when accessing sensitive data, so my information is protected.	M	User is prompted for a second factor after entering their password. Access to sensitive data is granted only if the second factor is correct. If the second factor is incorrect, the user cannot see the sensitive data and gets an error message. If the user does not complete the second factor, access is denied.	Sprint 10
14	Log out	As an Icelandic traveler, I want to log out easily so that no one else can use my device to view my data.	S	User can log out by a “Log out” button. User gets redirected to the log in view.	Sprint 8
15	Allergies	As an Icelandic traveler, I want my medical summary to include allergies so that I can inform foreign doctors quickly.	S	The allergies section is shown at the top of the summary when at least one allergy exists. Each allergy displays substance, reaction, severity and status. If no allergies are on record, the summary shows “No known allergies”.	Sprint 2

ID	Name	Description	Story Points	Acceptance Criteria	Sprint
16	Current prescription	As an Icelandic traveler, I want my medical summary to include current prescriptions so that doctors abroad know what I take.	S	Current prescriptions is shown clearly in the summary, with name, dose and frequency. If no prescriptions are on record, the summary shows “No prescriptions.”	Sprint 2
17	Chronic conditions	As an Icelandic traveler, I want my medical summary to include chronic conditions so that I doctors abroad know about them.	S	Chronic conditions are shown clearly in the summary. If no chronic conditions are on record, the summary shows “No chronic conditions.”	Sprint 2
18	Blood type	As an Icelandic traveler, I want my blood type on the medical summary so that doctors know it in emergencies.	S	Blood type is shown clearly in the summary.	Sprint 2
19	Vaccination history	As an Icelandic traveler, I want vaccination history included so that doctors abroad know my status.	S	Vaccination section is shown in summary, with vaccine name, dose and date.	Sprint 3
20	Emergency contact	As an Icelandic traveler, I want my emergency contacts list on my medical summary so that hospitals can reach them if needed.	S	An emergency contacts section is shown in summary, otherwise display “No emergency contacts on file”. Shows name, relationship, number and email.	Sprint 3
21	Primary doctor information	As an Icelandic traveler, I want my primary doctor’s contact info on the medical summary so that foreign doctors can contact them.	S	A primary doctor section is shown, otherwise “No primary doctor on file”. Name, role, organization, phone, and email.	Sprint 3

ID	Name	Description	Story Points	Acceptance Criteria	Sprint
22	Encrypted	As an Icelandic traveler, I want my offline downloaded file encrypted so that no one can read it if my phone is lost.	XL	The downloaded offline summary is stored encrypted. Opening the summary requires user authentication	Sprint 10
23	English	As an Icelandic traveler, I want to view my medical summary in English so that I can share it abroad.	M	User can select English from the language selector. Upon selection, the entire summary updates immediately to English. If translation fails, the app shows a clear error and keeps the last successfully displayed version.	Sprint 5
24	Default language	As an Icelandic traveler, I want to select my preferred default translation language so that I don't need to choose each time.	M	User can select a default translation language from a list of languages. When the user opens their medical summary, it automatically displays in the saved default language. User can temporarily switch languages in summary view, this does not change the default unless they save as default.	Sprint 9
25	Preview	As an Icelandic traveler, I want to preview the original and translated version so that I can confirm accuracy.	L	User can view original and translated side by side. If section can't be translated, the translated view shows "Not available – showing original."	Sprint 8
26	Read translated summary	As a doctor abroad, I want to see the translated medical summary so that I can treat the patient properly.	S	Doctor can open a translated summary.	Sprint 4

ID	Name	Description	Story Points	Acceptance Criteria	Sprint
27	QR code	As an Icelandic traveler, I want to generate a QR code of my summary so that doctors can scan it easily.	L	User can generate a QR code for the current summary with one action. The summary is shown in current selected language.	Sprint 9
28	PDF	As an Icelandic traveler, I want to share my medical summary as a PDF so that I can email it if needed.	L	User can export to PDF. PDF uses the current language. User can open the PDF offline. User can share via email or save locally. If export fails, shows a clear error message and retry option.	Sprint 6
29	Revoke shared files	As an Icelandic traveler, I want to revoke shared files after use so that my data isn't reused without my permission.	L	User can view a list of active shares with created time, expiry and last accessed. User can revoke any share and gets a success confirmation. Revoked links/QRs become immediately invalid. User can set an auto expire date.	Sprint 9
30	Notify when out-dated	As an Icelandic traveler, I want to be notified when my medical summary is out-dated so that I can update it.	M	User sees a date labeled "Last updated". User can refresh and the date changes.	Sprint 11
31	Multiple summaries	As an Icelandic traveler, I want to store multiple summaries so that I can travel with my children's data.	XL	User can add multiple profiles and see them in a profile switcher. Adding a child requires guardian verification. Each profile stores its own selected sections, language and offline copy.	Sprint 9
32	Fingerprint and face ID	As an Icelandic traveler, I want to be able to open my summary with fingerprint or face ID so that it's quick and secure.	L	User can open summary with face ID and fingerprint. After 3 failed attempts, they need to use PIN	Sprint 10

ID	Name	Description	Story Points	Acceptance Criteria	Sprint
33	PIN for offline version	As an Icelandic traveler, I want my offline summary to be encrypted with my device PIN so that only I can open it.	L	Offline summary is encrypted at rest. Opening the summary requires the device PIN. After 5 failed PIN attempts, access is temporarily locked for 5 minutes.	Sprint 10
34	Usage history	As an Icelandic traveler, I want to see when and where my summary was last accessed so that I can detect misuse.	L	User can open Access history showing date and time. Offline views are logged locally and sync to history when online.	Sprint 11
35	Delete offline version	As an Icelandic traveler, I want the option to delete my offline copy so that no traces remain on my device.	S	User can delete offline copy. On success, user sees “Offline copy deleted”. If deletion fails, app shows clear error and user can retry.	Sprint 8
36	Reminder to update	As an Icelandic traveler, I want reminders to update my summary before traveling so that I don’t forget.	M	User can set a trip with start date and enable Update reminder. App schedules a reminder 5 days before the trip.	Sprint 11
37	Notified when new data	As an Icelandic traveler, I want to be notified when new critical health data is available so that my summary stays relevant.	L	App detects critical changes. User receives a notification, showing what changed. User can refresh and the summary will update.	Sprint 11
38	Confirmation after exporting	As an Icelandic traveler, I want confirmation after exporting/sharing my summary so that I know it was successful.	S	After exporting PDF, app shows “Exported” with file name, size and save location. Failures show clear error with Retry option.	Sprint 8
39	Synced with my pages	As an Icelandic traveler, I want my summary synced with My Pages on Ísland.is so that I don’t need to re-enter data.	L	Data is pulled automatically from My pages in <i>Ísland.is</i> . First open fetches latest data and shows Last updated, time and date.	Sprint 5

ID	Name	Description	Story Points	Acceptance Criteria	Sprint
40	Simple look	As a healthcare professional accessing the medical summary for the first time, I want the entire look to be intuitive so I that I know what I need to do when seeing it.	M	Critical info is clear and on top, allergies, medication, conditions, blood type, vaccinations. Clear section headings with icons.	Sprint 4
41	Only important information	As a healthcare professional looking at the medical summary, I want important information only so that I don't waste too much time looking at it.	S	Summary shows critical info first; allergies, current medication, conditions, blood type, vaccinations, emergency contacts. Each section is clearly labeled.	Sprint 4
42	Information on one page	As a healthcare professional providing emergent care, I want the data to load in one page/ one screen format and uncluttered setting so that I can read the information comprehensively.	S	Summary. opens to a single screen view, with critical info; allergies, current medication, conditions, blood type, vaccinations, emergency contacts. Layout fits without scrolling.	Sprint 4
43	Guided tutorial	As an Icelandic traveler, I want a guided tutorial of the summary screen so that I can learn it quickly.	L	Tutorial auto appear on first use. Each step has a skip option.	Sprint 12
44	Dark mode	As an Icelandic traveler, I want dark mode support so that it's comfortable to read at night.	L	User can change to dark mode in system settings. All screens render in dark mode.	Sprint 12
45	Country auto detect	As an Icelandic traveler, I want country auto-detect to preselect language so the I don't have to switch manually.	XL	User can share location. Summary changes language according to the country the user is in.	Sprint 12

Chapter 4: Prototyping

4.1 Design phase

The design process of the medical summary extension builds on the preexisting design, color scheme, and motif of the *Ísland.is* app. The main goal was to create a user-centered design that adheres to the existing visual and functional standards, ensuring that users perceive the extension as a natural part of the main application. This approach allows for more authentic user reactions during testing, as participants interact with the feature in an environment that feels familiar and realistic.

The ideas that arose for what categories and information should be presented in order to create a medical summary were inspired by the following:

1. Interviews from Project B
2. Initial meeting with the stakeholders and the desire they expressed
3. Insights from the Ísland.is platform: particularly its support for electronic cross-border health services [1]
4. The International Patient Summary Implementation Guide which focuses on the minimal storage of confidential data [2].

Based on these influences, the team concluded that the medical summary should be structured around key categories used in electronic cross-border health records. They consisted of:

- Medical diagnoses
- Allergies and life-threatening conditions
- Current medications
- Implants
- Vaccinations
- Recent treatments and pregnancy information.

Therefore, after thoughtful consideration, in creating the medical summary page, the group decided to have categories in this structure:

- General information (e.g., contact and demographic details)
- Medical information (including allergies and vaccinations, organ donor)
- Medical diagnosis and condition (includes history and current cases)

- Current medications
- Implants
- Recent treatments and tests

In addition, the design incorporated a clear and intuitive navigation flow that aligns with the natural flow of the application, as shown in Figure C.1. Users begin on the home screen and access the *Health* section via the bottom navigation bar. From there, they can create a new medical summary or access previously created summaries. The process is supported by clear visual indicators and contextual text, such as "*Create Medical Summary*" and "*Previous Summaries*".

When creating a new summary, users are prompted to select which medical information should be included, followed by an option to select the language of the summary. This ensures accessibility for individuals who may need to share their information traveling abroad. A preview screen allows the user to verify all details before proceeding to export.

Since the goal of this prototype is to validate the main user flow for creating and translating a medical summary, full error and confirmation screens were not included at this stage. Because *Ísland.is* does not display many direct error messages in its current design, we added one example error state for the project purpose. The complete error and confirmation flows would be designed in a later sprints when the interaction logic is finalized.

4.1.1 Security and Data Handling

The security considerations for this prototype focused on understanding how data protection would differ depending on the export method. If the medical summary was added to the digital wallet, it would be encrypted and secured with biometric authentication. However, if it was exported as a PDF or printed, the data would be less secure. An additional option considered was to keep the medical summary stored only within the wallet feature for enhanced protection, with a limited access period for added security.

4.1.2 Added Features

A *Previous Medical Summary* feature was also proposed, allowing *Ísland.is* to recall an expired medical summary. This would reduce the number of steps required to recreate a new summary, especially when the user's medical information has not changed. The user could also modify the document's language without needing to update the underlying data.

The final design emphasizes simplicity, multilingual accessibility, and data security while maintaining adherence with the existing *Ísland.is* design framework. This ensures consistency and trustworthiness, crucial for health-related digital information.

4.2 User Testing

The goal of the user testing session was to evaluate how intuitive and user-friendly the Medical Summary feature is for users. Specifically, the objectives are to:

- Assess how easily users can locate, understand, and generate their medical summary.

- Identify pain points in navigation, labeling, and task flow.
- Gather qualitative feedback on design clarity and usefulness.

Our target group is Icelandic citizens and residents who use Ísland.is for personal or travel-related purposes. From this group, we aimed to evaluate five individuals with differing ages, genders and health literacies. In addition, we also tested users who were not Icelandic. The user testing sessions took place in person at Reykjavik University. Each session lasted approximately 25-30 minutes per participant. During the session, one team member acted as the facilitator, guiding participants through the test and asking follow-up questions when needed. Other team members took on the role of observers, documenting participants' interactions, comments, and any signs of confusion or hesitation.

Moreover, the testing was conducted using an interactive prototype of the Medical Summary feature, which was created in Figma. Participants were asked to complete a series of realistic tasks within the prototype. Before the session began, each participant was informed about the purpose of the test and asked to provide consent to participate. With permission, sessions were also recorded to support later analysis. All data collected was used solely for evaluating the usability and intuitiveness of the feature and to guide future design improvements. For the full user testing guide, see the Appendix A.

4.3 Happy Paths

The following four user tasks were designed to evaluate how intuitively users could navigate the prototype of Medical Summary feature within the Ísland.is Each task reflects a critical user action related to creating, managing, and exporting a medical summary. The group managed to create a total of five tasks that direct the user to the end of the process, with each task successfully moving the user towards the next step towards the end goal.

User Task 1: Create a medical summary

The user starts on the *Ísland.is* home screen, opens the **Health** section, and selects **Create Medical Summary**. They choose which medical information to include, such as allergies, medications, and medical conditions and so on, then generate the summary.

User Task 2: Make the Summary available in Spanish

After creating the summary, the user selects the **Language** option, chooses **Spanish**, and previews the translated version. This allows them to see how the medical summary would appear in another language.

User Task 3: Download the summary as a PDF

The user opens the **Export** menu, selects **Download PDF**, and previews the file before saving it. This lets them keep a copy of the medical summary for offline or printed use.

User Task 4: Add the summary to the wallet

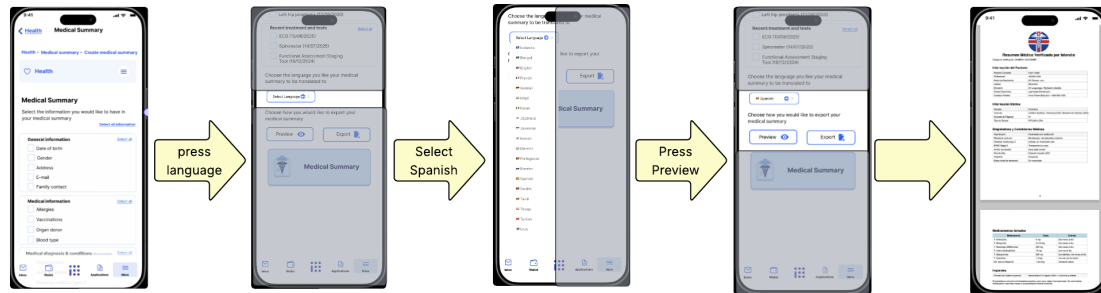
The user returns to the **Export** menu, selects **Add to Wallet**, and confirms the action. The medical summary is securely stored in the digital wallet for easy access later.

User task 5: View the summary in feature wallet

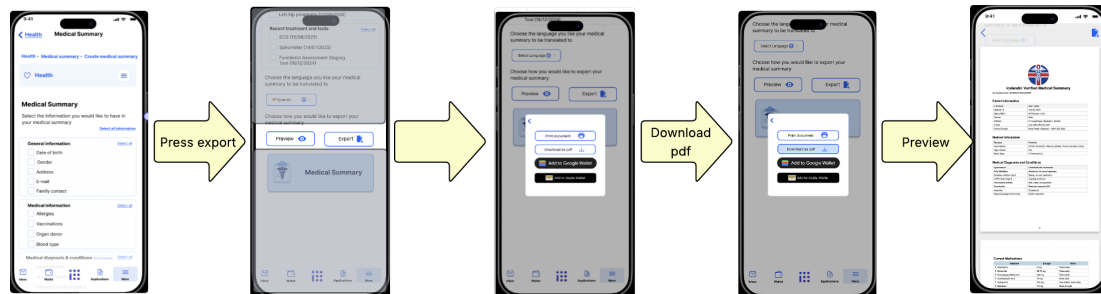
The user clicks the **Medical Summary**, which leads them to feature wallet. This lets them view the medical summary version and access date. The medical summary is then viewed by the link.



(a) User task 1: Create a medical summary

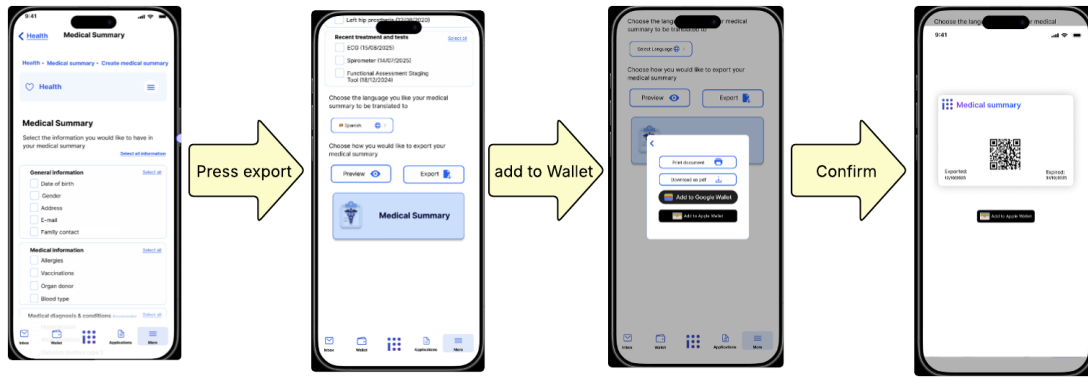


(b) User task 2: Make summary available in Spanish

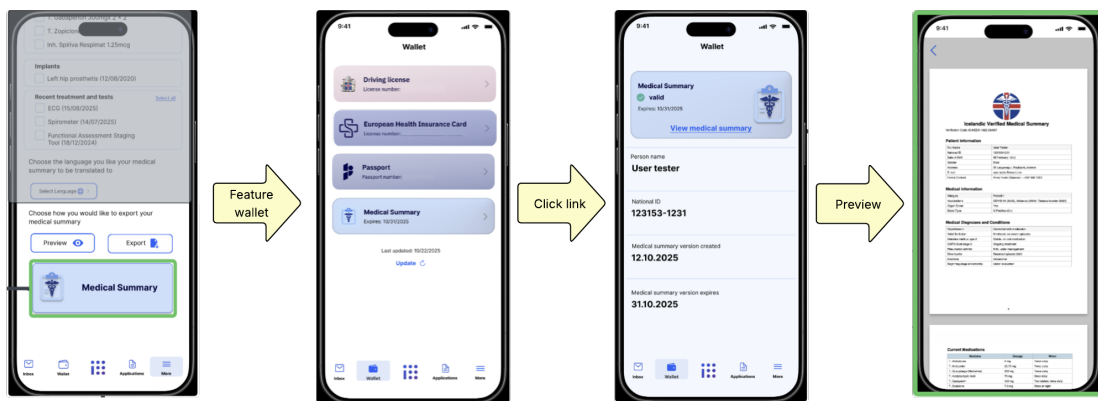


(c) User task 3: Download medical summary as a PDF

Figure 4.1: Prototyping of user tasks (part 1).



(d) User task 4: Add Medical Summary to your wallet



(e) User task 5: Feature wallet of medical summary

Figure 4.1: Prototyping of user tasks (part 2).

Chapter 5: User Testing Results

The team conducted five user tests and transcribed the interview questions (see Appendix B). During the user testing, one team member had the role of an observer, documenting the participants' interactions, comments, and any signs of confusion or hesitation. Here, you can see the results from each user for all the tasks. The tasks used during user testing can be found in Figure 4.1 and Section 4.3.

Task	Description
Task One	The user found the page for creating a medical summary very easily. He chose information that he wanted but found the export button quite confusing. He said that not everyone knows what "export" means and that it should say "Save" or "Download".
Task Two	The user attempted to locate an existing summary and translate it into Spanish, but soon realized that the prototype didn't support that feature. As a result, they created a new summary and set its language to Spanish.
Task Three	The user created a new summary and found it very easy to download as a PDF as he had already spotted that button.
Task Four	The user had already seen the "add to wallet" button and, therefore, had no trouble doing this.
Task Five	The user went straight to the right button and found it with no trouble.

Table 5.1: User One Results from User Testing

Task	Description
Task One	As a first-time app user, he had some initial trouble finding the section where you create the summary. After that, he had no issues creating it.
Task Two	Because he knew where to go from previous experience, the user completed the task without any issues.
Task Three	The user had already discovered the export as PDF function previously.
Task Four	The user had already discovered the export to wallet function previously.
Task Five	The user easily found out how to access the built-in wallet and pointed out that it's a very useful feature.

Table 5.2: User Two Results from User Testing

Task	Description
Task One	The user was unfamiliar with the <i>Ísland.is</i> app, yet managed to create a medical summary quite seamlessly. Some buttons were not working, which confused the user.
Task Two	The user was confused by how to get to the screen of creating a new medical summary. She missed the button and was also confused that the summary she just created was not showing up. Eventually she managed to create a Spanish summary but it took restarting the prototype.
Task Three	The user wanted to go back from the medical summary screen to the previous screen, that included the "Export PDF" function but the return buttons were not working. Eventually, through creating a new summary, the user found the button again and exported the PDF, yet was confused if the PDF was actually downloaded.
Task Four	This went smoothly. The save to wallet function was easily navigated to.
Task Five	The user quickly found the "view wallet" feature and looked at the medical summary in the wallet.

Table 5.3: User Three Results from User Testing

Task	Description
Task One	User followed the happy path but the prototype got stuck in the "No medical summary has been created" and needed to be restarted.
Task Two	User created a new summary but was confused by the lack of items in previous summaries.
Task Three	User followed the happy path but was confused by the fact that the prototype showed the english version instead of Spanish.
Task Four	User followed the happy path but accidentally pressed the preview while trying to de-select a language that was accidentally selected.
Task Five	User was confused by the fact that the wallet button does not work from the main screen.

Table 5.4: User Four Results from User Testing

Task	Description
Task One	The user was familiar with the <i>Ísland.is</i> layout. It took him little time to find the medical summary. He easily navigated through the flow which led him to the section of creating a medical summary.
Task Two	User easily followed the path and clicked a couple of languages before clicking Spanish, mainly out of interest.
Task Three	User clicked preview before clicking the export button and choosing the export file.
Task Four	User easily added the medical summary to his wallet.
Task Five	User clicked the wallet from the navigation bar from the same page as Task four and found the link easily.

Table 5.5: User Five Results from User Testing

User 1: The interviewee verbalized his thoughts throughout the tasks. One of the key points he mentioned was that the name and birth date should be required fields in the summary. During tests 2, 3, and 4, he repeatedly expressed a desire to access an already created summary and continue working on it. The facilitator explained that this feature is not yet available in the prototype but will be included in the final version. Overall, the user did not encounter any major difficulties, which suggests that the prototype is well-designed and easy to use.

User 2: The user initially experienced some difficulty navigating the application, as it was his first time using it and took a bit of time to understand the different sections. However, once familiar with the interface, he was able to perform tasks easily. The only complaint he had was about the placement of the "Create Summary" button. He would have preferred it to be above the previous summary instead of below it. Overall he liked the implemented features, in particular the possibility to add the summary to the in-app wallet.

User 3: As a first-time user, the user needed a moment to get acquainted with the app. After getting more familiar with the app, she found the app great and easy to use. Nevertheless, some

issues came up. The select buttons behaved in a way she found unintuitive and occasionally confusing. She mentioned that you should be able to "select all" for each section, which is currently not possible. Also, because the prototype's "return to previous page" button wasn't functional, she wasn't sure how to go back, when this seemed to be most efficient for some tasks. She was also uncertain whether previously generated PDFs are stored in-app or must be recreated each time, which caused some confusion. She mentioned that it would be great to be able to edit your already existing medical summary. Additionally, on the "My Medical Summary" page, she initially couldn't find the "Create Medical Summary" button because it was placed in a way that scrolling down was necessary.

User 4: As a frequent user of digital platforms but a self-described skeptic of government services, she found the feature technically appealing but had strong privacy concerns. She appreciated the convenience and usefulness of having her medical information summarized in one place, especially given her multiple health issues. The prototype itself was clear to her, though she noted that Figma prototypes limited functionality sometimes interfered with the testing process. She suggested making the language dropdown menu larger to improve usability. In general, she found the idea "very smart" and would likely use it despite her distrust of cybersecurity in government systems.

User 5: As a nurse living and working in Iceland, he immediately recognized the feature's practical value for both travelers and healthcare professionals. He emphasized that it could save time during patient intake and ensure more accurate treatment through accessible background information. He found the walkthrough easy to follow and the interface clear, particularly liking that the summary preview pops up directly on screen. She would feel safer and more at ease traveling abroad with access to such a summary, as it would provide legitimate and trustworthy medical information in emergencies.

Overall, the user testing provided clear insights into both usability and perceived value by the tested users. Across the five sessions, participants generally managed to complete the main tasks and recognized the practical benefits, such as for travel and efficient information sharing. At the same time, recurring issues emerged: several users struggled to locate key actions (especially creating a new summary and exporting it), were unsure whether previously created summaries could be viewed or edited, and encountered prototype-related limitations (e.g., non-functional buttons) that disrupted the app flow. Together, these findings show a need for clearer labels, better placement of primary actions, reliable navigation, and features that support continuity of use (viewing, editing, and reusing existing summaries). Specific recommended changes are as follows:

- Rename the "Export" action to a more familiar term (e.g., "Save" or "Download").
- Move the "Create Medical Summary" button higher on the page for better visibility.
- Add functionality to view, edit, and continue previously created summaries.
- Ensure all navigation controls (e.g., "Back") are reliable and consistently available.
- Provide a "Select All" option within each section to streamline selection.
- Clarify where exported PDFs are stored and how to access them inside the app.
- Increase the size and tap targets of dropdowns (e.g., language selection) for usability.
- Set certain fields as mandatory (e.g., name and date of birth) and change "address" to "home address", as well as "family contact" to "emergency contact".

This indicates that reducing the image size or moving the button higher would make it more visible. Overall, the usability issues found were minor and can be resolved.

5.1 Next steps

Based on the results of user testing and stakeholder feedback, we recommend continuing the development of the medical summary feature rather than dropping the idea. The concept was validated as useful, easy to understand, and aligned with stakeholder needs. However, we also found some issues that need to be fixed before moving forward. The next phase will focus on completing the prototype and running another round of user tests to confirm that the feature includes everything users need before moving into software development. Here is a list of next action items:

UX/UI Improvements:

- Add a clearer visual button for “Create Medical Summary”.
- Add small info icons to explain medical terms.
- Add a confirmation screen after creating the summary.
- Add a warning message before deleting the offline copy.
- Implement error screens for all key user actions.

Functionality to add:

- Add “Select All” button for each medical category.
- Add “Previous Medical Summary” option so users don’t start from scratch.
- Add automatic language detection + translation suggestion.
- Add automatic update when patient record changes.

Security and Privacy:

- Encrypt exported file with password or PIN.
- Add expiration time for offline stored summaries.
- Add warning if summary is shared on insecure channel.

User testing and validation:

- Conduct another usability test after implementing updates.
- Test with real doctors to confirm usability in hospital settings.
- Schedule third stakeholder meeting.

Chapter 6: Stakeholder meeting

On the 22nd of October at 12:00 noon, the team held its second meeting with the stakeholders from *Ísland.is*. The purpose was to provide a project update, confirm alignment with expectations, and gather feedback on the in-progress prototype.

During the meeting, the stakeholders were shown an early version of the prototype, which included the medical summary page and allowed users to manually select the information they wanted to include. At this stage, the prototype did not yet feature language selection or export options.

Key Feedback

- **Information selection:** Add a “*Select All*” option to simplify and speed up the process of choosing medical data.
- **Data security:** Require users to log in and authenticate before accessing or retrieving personal health information.
- **Offline access:** Consider enabling temporary data caching so users can access summaries offline after authentication.
- **Login mechanism:** Explore the use of *passkeys* or similar secure methods for authentication, especially for web access.

The feedback provided valuable input for refining the concept of the medical summary. Due to time constraints, the team prioritized enhancing the user interface and overall user experience when generating the summary. Efforts focused on implementing export options and evaluating user understanding, while features such as offline access and advanced login mechanisms were deprioritized because of limited time and the functional restrictions of the prototyping software.

Chapter 7: Conclusion

During this phase, the team transformed earlier research and planning into a functional prototype that captured the core value of the Medical Summary feature. The resulting minimum viable product enabled users to create, customize, and translate their medical summaries, laying a solid foundation for future development.

User testing confirmed that the concept is both practical and intuitive. While participants completed most tasks successfully, the tests revealed areas for improvement, including clearer labeling, easier access to existing summaries, and more intuitive navigation. These findings provide valuable direction for refining the design and emphasize the importance of focusing on what works best for the user.

The stakeholder meeting further underscored the feature's relevance, particularly in terms of data security, offline access, and simplified information selection. Although some enhancements could not be implemented within this phase, the feedback supports and informs the future development roadmap.

Overall, this phase showed that the medical summary feature is both practical and useful, with the prototype proving its potential and the feedback from users and stakeholders helping shape its next steps.

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Appendix A: User Testing Guide

Act 1: Friendly Welcome

Goal: Make participants comfortable and explain the session.

An Example Welcome:

“Hi! Thank you for joining this session. We’re testing a new feature that allows Icelandic citizens to download and translate their medical summaries from Ísland.is. We’re not testing you; we’re testing how clear and intuitive the app is. Please think aloud as you go through the tasks.”

Checklist:

- Confirm consent for participation and optional recording.
- Explain that the prototype is under development.
- Emphasize there are no right or wrong answers.

Act 2: General Context Questions

Goal: Understand the participant’s background and usage habits.

For Users:

1. First, can you tell me about your background, education, and current situation? primary school education, teacher, internship international
2. How often do you use the *Ísland.is* website or app?
3. Have you ever accessed your health information online (e.g., via Heilsuvera or My Pages)?
4. Have you ever needed to share medical information while traveling abroad?
5. How comfortable are you using digital government services?

Act 3: Introduction to the Prototype

Goal: Present the concept before performing tasks.

An Example Concept Introduction:

“This is a prototype of a new feature within the Ísland.is app that lets users access a Medical Summary. The summary includes key medical information, such as allergies, medications, and prior conditions, and can be downloaded or translated for travel.”

Participants are encouraged to explore briefly before beginning tasks.

Act 4: User Tasks

Goal: Observe how users interact with key features.

1. “Create a medical summary.”
2. “Try to make the summary available in Spanish.”
3. “Try to download your Medical Summary as a PDF.”
4. “Add your summary to your wallet.”
5. "View summary in your feature wallet."

Act 5: Debrief

Goal: Capture overall impressions and feedback.

1. What was your first impression of the feature?
2. Was there anything confusing or unclear?
3. How confident would you feel using this before traveling abroad?
4. If you could change one thing, what would it be?

Appendix B: User Tests

B.1 Interview 1

Act 2: General Context Questions

Q: First, can you tell me about your background, education, and current situation?

A: 46 years old from Iceland. Currently working as a customer success manager.

Q: How often do you use the Ísland.is website or app?

A: Once a week

Q: Have you ever accessed your health information online (e.g., via Heilsuvera or My Pages)?

A: Yes

Q: Have you ever needed to share medical information while traveling abroad?

A: Yes, my mother needed medical medical care abroad and I needed to help her get access to her medical information to share with the medical professionals abroad.

Q: How comfortable are you using digital government services?

A: Very

Act 5: Debrief

Q: What was your first impression of the feature?

A: It was simple and clear.

Q: Was there anything confusing or unclear?

A: Unclear how to open an existing medical summary

Q: How confident would you feel using this before traveling abroad?

A: Very

Q: If you could change one thing, what would it be?

A: I would like to have my most recent blood pressure tests, height and weight. I would like to be able to select all my medical documents within a certain time period and the document needs to show the time period. I think that the name and birth date should be mandatory. You should change address to home address and family contact to emergency contact.

B.2 Interview 2

Act 2: General Context Questions

Q: First, can you tell me about your background, education, and current situation?

A: I am a 24-year-old foreign student from Italy.

Q: How often do you use the Ísland.is website or app?

A: I have the Kennitala to access the app, but I have never used it.

Q: Have you ever accessed your health information online (e.g., via Heilsuvera or My Pages)?

A: No

Q: Have you ever needed to share medical information while traveling abroad?

A: I never needed to share information before, but since I always need to carry medicine with me due to health issues, I always have the prescription with me in case it's needed while traveling.

Q: How comfortable are you using digital government services?

A: I arrived in Iceland not so long ago, but I would probably say I'm comfortable.

Act 5: Debrief

Q: What was your first impression of the feature?

A: It is well implemented and seamlessly integrated with the existing Island.is app.

Q: Was there anything confusing or unclear?

A: Actually, I had some trouble initially understanding where I should go to create the summary, so I was slowed down while trying to figure out where to go. Maybe it's because I had never used the app before.

Q: How confident would you feel using this before traveling abroad?

A: I'm quite confident.

Q: If you could change one thing, what would it be?

A: As someone who has never used Island.is, I would probably put the medical summary button in a more accessible place since going to More > Health > Medical Summary was not intuitive. I would also put the "Create Medical Summary" button above the already created summaries and add the option to include emergency contacts in the summary.

B.3 Interview 3

Act 2: General Context Questions

Q: First, can you tell me about your background, education, and current situation?

A: So, I am German and 24 years old. I study primary school education, so I am a teacher and currently I am doing an internship at an international school here in Iceland.

Q: How often do you use the Ísland.is website or app?

A: Never.

Q: Have you ever accessed your health information online (e.g., via Heilsuvera or My Pages)?

A: No.

Q: Have you ever needed to share medical information while traveling abroad?

A: Yes. When I was in Canada, I was sick and had to go to the doctor and then I had to send the receipts of my doctors to my doctor back home.

Q: How comfortable are you using digital government services?

A: Probably depends on the government, but in general comfortable.

Act 5: Debrief

Q: What was your first impression of the feature?

A: So, in general, the idea, I really like. I think it's very useful for like traveling, to have like all your information because you never know when you might need them. I think the idea is great. It was a bit weird doing it on the computer when it looks like a phone but I think, obviously in the end it would be an app. So, in general, I think that how it looked is nice and it is very clear... because there are not like many options, there are few options you can choose, and it's like very straightforward how you create the medical summary. So, yeah, I think it's a bit, I don't know, maybe it was me, but I feel like every time you wanna change something about the medical summary, you have to create a new one, which can be maybe a bit annoying. But also, it is not that many buttons, so it's not, like, taking you forever to create a new one.

Q: Was there anything confusing or unclear?

A: Uhm, yeah, I think, at first, I wasn't sure like if I wanna change something about the medical summary, do I have to create a new one, or can I change the one I already did. So, I think that is something that you have to figure out, but, then, once you know, you just need to create another one, it is very straightforward. And I think it takes some time to figure out where is what, but since it is not that many, like buttons and options, I think it's like, you work yourself into the app for like a couple of minutes and then it seems... great.

Q: How confident would you feel using this before traveling abroad?

A: I think I would feel comfortable. Like I said, it would be nice to have like all of the information summarized, and it's very helpful for like traveling. So, if something like this exists or would exist, I would probably use it.

Q: If you could change one thing, what would it be?

The thing with "select all" or "unselect all". So that you can just ... that it would work straightforward. But besides that, I don't know ... looks nice.

B.4 Interview 4

Act 2: General Context Questions

Q: First, can you tell me about your background, education, and current situation?

A: I'm currently in my last year in my BSc degree in Computer Science, and I work in public outreach in a cybersecurity research lab and a Lead TA.

Q: How often do you use the Ísland.is website or app?

A: Every Two months.

Q: Have you ever accessed your health information online (e.g., via Heilsuvera or My Pages)?

A: Yes.

Q: Have you ever needed to share medical information while traveling abroad?

A: Yes, I had to go to the hospital in Bulgaria in 2021.

Q: How comfortable are you using digital government services?

A: I use them because I have to, I don't trust them.

Act 5: Debrief

Q: What was your first impression of the feature?

A: 1. It's very smart, especially since I have way to many medical problems, so it would be nice to have this.

Q: Was there anything confusing or unclear?

A: 2. No, just Figma being a bitch.

Q: How confident would you feel using this before traveling abroad?

A: I would feel very confident in usability, however I am very untrusting with everything related to cybersecurity and especially with government stuff, so I would use it but hate to have to use it.(Paranoia)

Q: If you could change one thing, what would it be?

A: Make the languages dropdown menu larger.

B.5 Interview 5

Act 2: General Context Questions

Q: First, can you tell me about your background, education, and current situation?

A: I am Filipino who came to Iceland 8 years ago. I graduated as Nurse and I currently live here in Iceland and work as a Nurse.

Q: How often do you use the Ísland.is website or app?

A: Not quite often, I just use it to check emails from the government and to record my milage every month since I drive an electric car.

Q: Have you ever accessed your health information online (e.g., via Heilsuvera or My Pages)?

A: Yeah I always use Heilsuvera when I have to check something medical related.

Q: Have you ever needed to share medical information while traveling abroad?

A: No, I haven't needed to do that yet.

Q: How comfortable are you using digital government services?

A: I am semi-comfortable. I always need to use authentication, so I always need to put it everytime and it's annoying. I don't get why it needs to be done for every email, but if it was an important document unlike the milage documentation.

Act 5: Debrief

Q: What was your first impression of the feature?

A:The medical summary, I think it is very useful especially for the traveler, essential and for the healthcare practitioner working abroad, so they have the background of the patient. So they have the information. It is much easier time saving, healthcare practitioner could cater other patient who are waiting. I think the dates information is great and improves the medical history or note taking when doing the initial assessment of the patient.

Q: Was there anything confusing or unclear?

A: I think no, the walkthrough is kinda okay, it is easy to understand and not confusing. The information is there which makes sense. I like the feature when you view the summary it will pop up.

Q: How confident would you feel using this before traveling abroad?

A: I would feel much more safer..at ease, if ever what happens to me, there will be a legit information about my background and history that they will base on. Some medical investigation will be reduced.

Appendix C: Userflow

